

Requirements, workflows & UAT

The analysis behind a RAG knowledge system and an automated prospecting workflow: who needed what decision, how the operating process changed, what the system must do, and how release readiness is checked.

01 · RAG KNOWLEDGE SYSTEM
Turn institutional knowledge into grounded decision support
Make repeated support resolutions and contract language searchable, source-linked, and usable inside the decision workflow.

02 · PROSPECTING OPERATIONS
From manual lookups to mapped candidate coverage
Replace one-at-a-time research with a governed search, qualification, pipeline, and cost-control process.

My role: stakeholder discovery, problem framing, current/future workflow design, requirements and acceptance criteria, data ownership, implementation, test design, rollout, and measurement planning.

Stakeholder decision map

STAKEHOLDER	DECISION OR JOB	OPERATING NEED	DESIGN RESPONSE
Support agent	Resolve a live question while the customer is waiting.	Find the applicable prior resolution without depending on a specific experienced agent.	Natural-language retrieval in the team's established messaging channel, with the source passage available for verification.
Support manager	Staff more flexibly and reduce escalation dependency.	Give newer and night-shift representatives reliable access to institutional knowledge.	Source-grounded answers, low-friction access, escalation behavior, and lookup/adoption measures.
Knowledge owner	Preserve jurisdiction- and case-specific context.	Exact codes, dates, amounts, categories, and policy versions matter alongside semantic similarity.	Hybrid search, enriched metadata, source links, correction ownership, and a failure-review loop.
ATM business owner	Choose a scalable acquisition motion with visible cost.	Increase market coverage while keeping API use and downstream workload understandable.	Pre-run estimate, explicit paid action, caps, cache, stop control, and campaign economics.
Business-development user	Find, qualify, and advance prospects without duplicate work.	See coverage, existing sites, pipeline status, uncertainty, and next action in one place.	Adaptive search, tri-state qualification, conservative deduplication, history, revisit logic, and export.
Data / security owner	Control approved sources, access, lifecycle, and incident response.	Keep confidential content within the authorized operating boundary.	Source inventory, role access, local-first processing, minimized message output, retention ownership, and release checkpoints.

Discovery frame

QUESTION	RAG KNOWLEDGE SYSTEM	ATM PROSPECTING
What decision changes?	Which prior resolution or clause applies, and which source should the user verify?	Is this candidate worth the next action, and what evidence supports that decision?
What is the unit of work?	A verified answer, correction, or escalation.	A unique candidate with a current qualification and funnel state.
What creates risk?	Unsupported answers, stale sources, missed exact terms, or content leaving its approved boundary.	Incomplete coverage, uncontrolled calls, duplicate outreach, false exclusion, or stale pipeline state.
How is value read?	Time to verified resolution, escalation burden, answer support, and adoption.	Coverage, analyst capacity, qualified output, campaign conversion, and total acquisition cost.

Confidentiality: Public examples use reconstructed or non-confidential records. Employer source code and confidential records are excluded.

From current state to operating workflow

RAG knowledge system

BEFORE	Search long threads or recordings, guess the right terms, and interrupt an experienced agent when the answer is unclear.	→	Ask in plain language, receive a source-grounded answer, verify the cited passage, then answer or escalate with context.
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STEP	OWNER	ACTION	SYSTEM BEHAVIOR	CONTROL / EXCEPTION
01	Data owner	Approves source, purpose, access, and lifecycle.	Records the source, version, and owner.	Unapproved source stays outside the corpus.
02	Knowledge owner	Defines authoritative content and conflict rules.	Transcribes/OCRs, cleans, enriches, chunks, and indexes content.	Low-quality extraction routes to review.
03	Support agent	Asks a natural-language question in the established work channel.	Combines semantic and exact-term search, then builds a cited response.	Weak or conflicting evidence triggers clarification or escalation.
04	Support agent	Checks the cited passage and acts on the answer.	Preserves document, page, section, and source context.	Missing citation means the answer is not complete.
05	Manager / owner	Reviews corrections, use patterns, and recurring failures.	Adds new tickets and calls through the live ingestion pipeline.	Correction becomes a test case and source-maintenance action.

ATM prospecting workflow

BEFORE	Search maps one location at a time, copy listings into Excel, and rely on memory to avoid missed areas or duplicate review.	→	Scope a market, forecast usage, search adaptively, qualify and dedupe candidates, record decisions, and advance a governed pipeline.
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STEP	OWNER	ACTION	SYSTEM BEHAVIOR	CONTROL / EXCEPTION
01	Owner / BD	Sets the market, search shape, categories, and budget.	Shows expected API usage and cost before a search begins.	No explicit start means no paid search.
02	BD operator	Starts and monitors the search.	Searches restricted geometry, subdivides dense areas, and reuses valid cache.	Budget, burst, daily, lifetime, and kill controls can stop calls.
03	BD operator	Reviews candidate evidence and exceptions.	Normalizes records, overlays sites/pipeline, deduplicates conservatively, and preserves unknown fields.	Missing evidence routes to verify; it does not become a false "no."
04	BD owner	Records qualification, reason, stage, owner, and next action.	Persists history, revisit dates, and campaign membership.	Bulk updates cannot erase a later stage or human decision.
05	Owner / analyst	Exports current records and reviews market/campaign performance.	Reconciles candidate counts, workflow cost, and closed-campaign outcomes.	Coverage and outcome metrics remain tied to their scope and period.

Requirements and acceptance criteria

RAG knowledge system

ID	REQUIREMENT	ACCEPTANCE CRITERION	OWNER	VERIFICATION
RAG-01	Approved knowledge sources	Every indexed source has an owner, business purpose, access rule, and update path.	Data / knowledge owner	Operating control
RAG-02	Hybrid retrieval	Paraphrased and exact-term questions draw from fused vector and keyword candidates.	Solution owner	Implemented · source
RAG-03	Source-linked answer	Each answer citation resolves to the retrieved document and supporting passage.	User / knowledge owner	8 / 8 checks · test
RAG-04	Clarify or escalate	When evidence is weak, conflicting, or out of scope, the workflow requests context or routes human review.	Business owner / agent	Expand in UAT
RAG-05	Document ingestion	PDF, DOCX, TXT, scans, and filled-form overlays retain usable text and source metadata.	Solution owner	Implemented · source
RAG-06	Protected processing	Deployment mode follows the approved data boundary; supported banking patterns are masked before external-capable processing.	Security / solution owner	Tested · test
RAG-07	Feedback and maintenance	Corrections and escalations are categorized, assigned, and added to the regression set after resolution.	Support manager	Next iteration

ATM prospecting workflow

ID	REQUIREMENT	ACCEPTANCE CRITERION	OWNER	VERIFICATION
ATM-01	User-controlled spend	Paid requests begin only after an explicit action; projected usage is visible first and enforced limits stop additional calls.	Business owner / operator	Implemented + tested
ATM-02	Adaptive coverage	Dense areas are subdivided only when needed; exact repeats use cached child coverage without new upstream calls.	Solution owner	Tested · search tests
ATM-03	Preserve uncertainty	Missing provider evidence remains unknown and is routed to review rather than treated as a negative fact.	BD / data owner	Implemented + tested
ATM-04	Unique candidate record	Results merge only under conservative same-location rules; adjacent businesses stay distinct.	BD / solution owner	Tested · dedupe tests
ATM-05	Governed pipeline state	Status, reason, owner, stage, history, revisit date, and campaign remain attached to the candidate; bulk updates preserve later progress.	BD process owner	Tested · workflow tests
ATM-06	Operational handoff	Exports carry qualification and current pipeline state, not just raw provider listings.	BD / operations	Implemented
ATM-07	Outcome measurement	Closed campaigns record mailed volume, responses, signed and productive sites, source/market, and total acquisition cost.	Business owner / analyst	Outcome plan ready

Traceability principle: requirements describe observable business behavior. Code and automated tests support technical acceptance; representative users own business UAT and release acceptance.

Data and operating controls

SOURCE	BUSINESS USE	OWNER	QUALITY RULE	OPERATING CONTROL
Support email threads	Prior questions, resolutions, exceptions, dates, and categories.	Support / data owner	Remove duplicates and boilerplate; retain the original reference and policy context.	Approved role access, lifecycle mapping, source version, correction path.
Recorded support calls	Resolutions that may never have been written into a ticket.	Support / data owner	Transcribe, check low-confidence audio, and distill the core question and resolution.	Authorized local storage, role-based access, and retention ownership.
Enriched knowledge records	Searchable question, resolution, state, category, date, and source link.	Knowledge owner	Keep generated metadata subordinate to the source; sample by category and review corrections.	Live ingestion, stale-source review, failure backlog, and re-index procedure.
Contract documents	Clause retrieval, supporting passage, and derived deadline calculations.	Contract / data owner	OCR scans and overlays; keep stated facts distinct from derived values.	Local-first default, source citations, masked supported banking patterns.
Google Places results	Candidate identity, address, category, hours, contact, and search coverage.	Provider + business data owner	Normalize fields, preserve missingness, dedupe conservatively, and timestamp searches.	Restricted geometry, caching, spend limits, provider/source review.
Site and pipeline records	Existing coverage, candidate qualification, ownership, history, campaigns, and revisit dates.	BD process owner	One governed record per candidate; prevent stale or backward bulk changes.	Audit history, reason codes, monotonic stages, controlled export.

Accountability model

ROLE	OWNS	APPROVES	RECEIVES / REVIEWS
Business process owner	Scope, users, decision rights, KPI, funding, rollout and rollback.	Release and operating policy.	Adoption, outcome, and exception review.
Data / knowledge owner	Source authority, conflict resolution, quality, lifecycle, and correction.	Corpus and field use.	Stale-source and quality reports.
Representative user	Scenario feedback, exception discovery, and UAT execution.	Workflow usability and critical-path acceptance.	Release notes and support path.
Security / IT	Access, device/channel configuration, processor boundary, logs, incidents, recovery.	Deployment control set.	Access reviews, incidents, and control changes.
Solution owner / analyst	Requirements traceability, implementation, tests, monitoring, defects, and change control.	Technical release candidate.	UAT defects, production signals, and owner decisions.

Deployment control checklist

✓ **Access:** named roles, least privilege, periodic membership review, credential rotation, and device expectations.

✓ **Content:** approved source classes, local-first processing, minimized message output, and a defined escalation path.

✓ **Lifecycle:** raw, derived, index, log, cache, and backup records follow named retention and deletion owners.

✓ **Operations:** monitoring, cost/error thresholds, backup/recovery, incident response, change record, and rollback path.

Messaging channel: Telegram was the company-standard communication platform, so the support assistant met agents inside their existing workflow. Channel access, retention, minimized output, and incident handling belong in the same operating controls as the rest of the system.

UAT and release readiness

5 / 5

Held-back support questions passed after ten tuning questions.

22 / 22

Expected contracts retrieved; 20 / 22 expected passages and 8 / 8 citation checks.

57 tests

Search, cache, spend, qualification, history, revisit, and pipeline controls.

Representative UAT scenarios

Scenario	Expected Business Result	Current Signal	Release Use
Support question uses different wording from the source.	Retrieve the applicable prior resolution and show the supporting passage.	Pilot passed	Expand across common ticket categories and agent seniority.
Question depends on an exact code, state, amount, or clause.	Hybrid search keeps the exact business term in the retrieved evidence.	Validated	Include high-risk and jurisdiction-sensitive cases in owner UAT.
No source adequately supports an answer.	Ask for clarification or route a human escalation with the available context.	UAT focus	Lock the owner-approved abstention and escalation policy.
A map search covers a dense market.	Increase coverage adaptively while staying inside forecast and enforced call limits.	Validated	Run representative market, density, and budget scenarios.
Provider metadata is incomplete or candidates overlap.	Preserve unknown fields for review and avoid merging adjacent businesses.	Validated	Sample exception decisions with a BD owner.
An import carries older pipeline state.	Keep current progress, record the attempted change, and preserve ownership/history.	Validated	Complete one end-to-end campaign handoff and export.

Decision log

Decision	Why	Tradeoff Managed	Revisit Trigger
Hybrid retrieval	Questions contain both paraphrase and exact business terms.	More components, fewer single-retriever blind spots.	Error analysis shows one path adds no value.
Source-first answer design	Users need to act on the answer, not merely read fluent text.	One verification step in exchange for traceability and correction.	User behavior or policy changes the required verification level.
Existing work channel	Reach distributed agents without a new tool or training burden.	Channel configuration joins the deployment control set.	Company platform or security policy changes.
Paid Places provider	The operating workflow needs business-listing coverage and fields suitable for qualification.	Provider dependency managed through forecast, cache, limits, and stop control.	Coverage, cost, terms, or field availability changes.
Tri-state qualification	Missing listing data cannot prove a feature is absent.	More human review, fewer false exclusions.	An authoritative field source closes the missingness gap.
Campaign outcome tracking	Search activity matters only when it connects to business-development results.	Additional operating discipline creates usable acquisition economics.	KPI or sales motion changes.

Release checkpoints

✓ Business owner confirms scope, target users, decision rights, KPI, rollout, and rollback.

✓ Representative users complete critical scenarios; defects have an owner and disposition.

✓ Data and security owners confirm sources, access, lifecycle, channel, and incident controls.

✓ Solution owner confirms monitoring, cost/error thresholds, backup, recovery, and support path.

Measurement after release: track verified lookup time, escalation and adoption for the knowledge workflow; track unique candidates, qualification completion, mailed volume, responses, signed and productive sites, cycle time, and total acquisition cost for prospecting.